

Systematic Closure of Minimal First-Principles Routes to Dark Energy in Einstein-Cartan-Holst Gravity

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Abstract

Einstein-Cartan-Holst (ECH) gravity—the Palatini formulation of general relativity supplemented by the Holst term with Barbero-Immirzi parameter γ —is a natural candidate framework for geometric dark energy. The minimal ECH+Dirac model generates a parity-odd vacuum operator through spin-torsion coupling, and a phenomenological scaling argument relates its inflationary dilution to the observed dark-energy density. However, the late-time equation of state $w = -1$ must be assumed, not derived: it requires showing that integrating out spin degrees of freedom generates an IR-constant term in the effective action. We report four systematic attempts to close this gap—two computational and two structural—together with their negative outcomes:

- (i) **Track B (condensate route):** Fierz rearrangement of the torsion-induced four-fermion interaction shows that the scalar/pseudoscalar channel coupling $G_{\text{SP}} \propto (\gamma^2 - 1)/(\gamma^2 + 1)$ is repulsive at $\gamma < 1$. Even when attractive ($\gamma > 1$), the coupling is $\sim 175\times$ below the NJL critical coupling, and gravitational catalysis is exponentially suppressed. Gate 1 (condensate existence) fails; Track B closes.
- (ii) **Branch G v1 (strict one-loop effective action):** After exact torsion elimination, all finite- γ dependence resides in the four-fermion term $S_{4\text{f}}$, which is quartic in fermion fields and does not contribute to the strict one-loop fermion determinant. The Dirac operator is the standard Levi-Civita operator with γ -independent Seeley-DeWitt coefficients. Branch G v1 closes at Phase 1 (no novel Holst content at this order).
- (iii) **Route T1 (dynamical Immirzi field):** Promoting the Barbero-Immirzi parameter $\gamma \rightarrow \theta(x)$ to a dynamical pseudoscalar field. Torsion remains algebraic, but after torsion elimination the reduced action is identical to GR + standard axion-like particle (ALP). The four-fermion interaction is θ -independent; no mechanism generates a potential $V(\theta) \sim \Lambda$; cosmological dynamics gives $w = +1$ (stiff matter), not $w = -1$. Route T1 closes: the dynamical Immirzi field has no novel gravitational content beyond a standard ALP.
- (iv) **Route S1 (parity-sensitive CMB phenomenology):** Assessment of whether the framework’s parity-odd operator maps to distinctive birefringence observables. The minimal ECH+Dirac model has no photon-polarization coupling; any assumed coupling produces constant- β birefringence identical to generic ALP phenomenology with two free parameters for one observable. Route S1 closes: no distinctive parity signal exists in the minimal model.

These closures share a common structural origin: in the minimal ECH+Dirac model, torsion is algebraic (non-propagating) and its elimination removes all Holst-sector content from the reduced action. The Nieh-Yan topological density carries no local dynamical information that survives this elimination. The resulting physics is standard general relativity plus fermions

plus a constrained four-fermion contact interaction—insufficient for late-time vacuum energy or distinctive observational signatures. The phenomenological framework [1] survives as a motivated ansatz; progress toward a first-principles derivation requires physics beyond the minimal model class. All computations are symbolic or semi-analytic (Python/SymPy) and are fully reproducible.

1 Introduction

The cosmological constant problem—why $\rho_\Lambda \sim (2.3 \text{ meV})^4 \sim 10^{-122} M_{\text{Pl}}^4$ —motivates the search for geometric mechanisms that generate or explain the observed vacuum energy from the structure of spacetime itself. Einstein-Cartan-Holst (ECH) gravity is a natural candidate: its Holst term introduces a parity-odd coupling controlled by the Barbero-Immirzi parameter γ , well-motivated by loop quantum gravity [2, 8]. The spin-torsion interaction generates four-fermion contact terms and a parity-violating vacuum operator, suggesting a possible geometric origin for dark energy.

A phenomenological framework based on this idea [1] constructs a dark-energy density $\rho_\Lambda = \Xi M_{\text{Pl}}^4$ through inflationary dilution of the parity-odd operator, reducing fine-tuning from 10^{120} to $\sim 10^5$. However, the framework contains a logical gap: the operator $\langle \epsilon^{abcd} K_{ab} R_{cd} \rangle$ is sourced by the fermion spin density, which vanishes at late times. For the residual to persist as a genuine vacuum energy with $w = -1$, one must show that integrating out the spin degrees of freedom generates an IR-constant term in the effective action.

This paper reports a structured program to close that gap. Four routes were tested—two computational, one structural extension, and one phenomenological assessment. All four fail. These are clean negative results: they narrow the landscape of viable derivation strategies, close the most natural minimal-model candidates, and establish concrete structural requirements for any next-generation geometric dark-energy mechanism.

Scope and limitations

This paper reports results for the *minimal* ECH+Dirac model (Einstein-Cartan gravity with Holst term, minimally coupled to N_f Dirac fermions, at the LQG Barbero-Immirzi value $\gamma = 0.274$), together with the simplest structural extension (promoting γ to a dynamical field). Routes involving non-minimal couplings, higher-loop effects, non-perturbative dynamics, graviton-loop contributions, genuinely new symmetry structures, or different values of γ are not tested and remain open. The negative results reported here apply only at the stated approximation orders and model classes.

2 Context and Motivation

The phenomenological framework of Ref. [1] treats $w = -1$ as an explicit assumption, stating: “We therefore treat $w = -1$ as a phenomenological assumption [...] and flag the derivation of the late-time IR effective action as an important open problem.” The present paper constitutes a systematic test of whether the minimal microscopic theory delivers such a term.

The question is not merely whether a specific calculation succeeds, but whether the *model class*—minimal ECH gravity with algebraic torsion and standard Dirac coupling—has the structural capacity to generate late-time vacuum physics. The four routes tested here cover the most natural candidates within this model class:

- condensation of the torsion-induced four-fermion interaction,

- the one-loop fermion effective action,
- promotion of γ to a dynamical field, and
- mapping the parity-odd operator to distinctive observables.

The simultaneous failure of all four points to a structural limitation of the model class itself, not merely to the failure of individual calculations.

3 Minimal Model and Tested Question

3.1 Microscopic action

The starting point is the Einstein-Cartan-Holst action minimally coupled to N_f Dirac fermions:

$$S = S_{\text{grav}}[e, \omega] + S_{\text{ferm}}[e, \omega, \psi], \quad (1)$$

$$S_{\text{grav}} = \frac{M_{\text{Pl}}^2}{2} \int [\varepsilon_{IJKL} + \frac{2}{\gamma} \eta_{I[K} \eta_{L]J}] e^I \wedge e^J \wedge F^{KL}[\omega], \quad (2)$$

$$S_{\text{ferm}} = \int d^4x |e| [\bar{\psi}_i i\gamma^\mu D_\mu \psi_i - m_i \bar{\psi}_i \psi_i], \quad (3)$$

with $D_\mu = \partial_\mu + \frac{1}{4}\omega_\mu^{IJ}\gamma_I\gamma_J$, metric signature $(+, -, -, -)$, $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ with $(\gamma^5)^2 = 1$, and $\kappa^2 = 8\pi G = M_{\text{Pl}}^{-2}$. The Barbero-Immirzi parameter γ is a fixed constant. No bare cosmological constant is included in the starting action.

3.2 The central question

Does integrating out fermion fields in the action (1) generate a late-time vacuum term with $w \approx -1$ that is not merely a restatement of bare- Λ renormalization and that depends nontrivially on finite γ ?

Four routes were tested, corresponding to different approximation strategies and extensions:

- **Track B:** Test whether a pseudoscalar fermion condensate $\Phi = \langle \bar{\psi} i\gamma^5 \psi \rangle$ forms via the NJL mechanism from the torsion-induced four-fermion interaction and persists as late-time vacuum energy.
- **Branch G v1:** Compute the strict one-loop gravitational effective action $\Gamma_{\text{eff}}[e]$ after torsion elimination and fermion integration, and test whether it contains a γ -dependent vacuum-like term beyond standard cosmological-constant renormalization.
- **Route T1:** Promote the Barbero-Immirzi parameter to a dynamical pseudoscalar field $\gamma \rightarrow \theta(x)$ and assess whether the resulting theory contains novel gravitational content (non-ALP terms, θ -dependent four-fermion structure, or a dynamically generated vacuum potential).
- **Route S1:** Assess whether the framework's parity-odd operator $\varepsilon^{abcd}K_{ab}R_{cd}$ maps to distinctive, testable birefringence observables that are not reproducible by generic axion-like-particle models.

4 Track B: Condensate Route

4.1 Setup

The spin connection ω_μ^{IJ} is non-propagating in EC gravity; its equation of motion is algebraic. Solving it exactly yields the reduced action [2, 3]

$$S_{\text{reduced}}[e, \psi] = S_{\text{EH}}[e] + \int d^4x \sqrt{-g} [\bar{\psi} i \gamma^\mu \overset{\circ}{\nabla}_\mu \psi - m \bar{\psi} \psi] + S_{4\text{f}}[e, \psi; \gamma], \quad (4)$$

where $\overset{\circ}{\nabla}$ uses the torsion-free Levi-Civita connection and $S_{4\text{f}}$ is the torsion-induced four-fermion interaction.

Perfect-square structure. The four-fermion term has a constrained form discovered during Computation 1 [18]:

$$\mathcal{L}_{4\text{f}} = -G_{\text{eff}} \left(\bar{\psi} \gamma^\mu \gamma^5 \psi + \frac{1}{\gamma} \bar{\psi} \gamma^\mu \psi \right)^2, \quad G_{\text{eff}} = \frac{3\kappa^2}{16} \frac{\gamma^2}{\gamma^2 + 1}. \quad (5)$$

This is a perfect square of a single combined current $J^\mu = J_{(A)}^\mu + (1/\gamma) J_{(V)}^\mu$. The individual couplings G_V , G_A , G_{VA} are *not* independent parameters; they are constrained by this squared-current structure.

At $\gamma = 0.274$, the vector admixture $1/\gamma \approx 3.65$ is $\mathcal{O}(1)$ —the vector sector dominates over the axial sector.

4.2 Gate structure

Track B was organized around three quantitative gates, defined and frozen before computation began [16]:

Gate 1 (Condensate existence)

The renormalized effective potential $V_{\text{eff}}(\sigma, \pi)$ admits a local minimum at (σ^*, π^*) with $\pi^* \neq 0$ and positive-definite Hessian.

Gate 2 (Vacuum persistence)

The minimum persists when curvature $R \rightarrow 0$ and spin source $s \rightarrow 0$.

Gate 3 (Vacuum-like stress-energy)

The condensate contribution yields $\rho_\Phi > 0$ and $|1 + w| < 10^{-2}$, with scheme-independent existence/sign.

Gate 1 failed; Gates 2 and 3 were not reached.

4.3 Algebraic and numerical results

Result 1: The scalar/pseudoscalar channel is repulsive at $\gamma = 0.274$. Fierz rearrangement of the four-fermion interaction (5) decomposes it into standard NJL channels [19, 20]. The effective coupling in the scalar/pseudoscalar sector is

$$G_{\text{SP}} = \frac{3\kappa^2}{16} \frac{\gamma^2 - 1}{\gamma^2 + 1}. \quad (6)$$

This is an exact algebraic result. The sign changes at $\gamma = 1$:

- $\gamma > 1$: attractive (condensation possible in principle),
- $\gamma = 1$: decoupled,
- $\gamma < 1$: **repulsive** (condensation impossible via NJL).

At $\gamma = 0.274 < 1$, the scalar/pseudoscalar channel is repulsive. The NJL gap equation has no nontrivial solution for any repulsive coupling, regardless of curvature or fermion content.

Result 2: Even with the correct sign, the coupling is $\sim 175\times$ below NJL criticality. The NJL critical coupling for condensation in proper-time regularization is $G_{\text{crit}} = 2\pi^2/(N_c\Lambda^2)$ [22]. With gravitational coupling $G \sim \kappa^2 \sim 1/M_{\text{Pl}}^2$ and cutoff $\Lambda \sim M_{\text{Pl}}$:

$$\left. \frac{G_{\text{SP}}}{G_{\text{crit}}} \right|_{\gamma=2.0} \approx 0.006. \quad (7)$$

The gravitational four-fermion coupling is subcritical by two orders of magnitude, even when the sign is attractive.

Result 3: Gravitational catalysis is exponentially suppressed. Gravitational catalysis [5] lowers the threshold for *already-attractive* channels but cannot flip the sign of a repulsive channel. For the attractive case ($\gamma > 1$) at Planck curvature $R \sim M_{\text{Pl}}^2$ [22]:

$$M_* \sim \Lambda \exp(-\text{const}/(G_{\text{SP}} \times R)) \sim \exp(-2100) \approx 10^{-914} M_{\text{Pl}}. \quad (8)$$

The corresponding vacuum energy $V_{\text{vac}} \sim M_*^4 \sim 10^{-3658} M_{\text{Pl}}^4$ is smaller than the observed $\rho_\Lambda \sim 10^{-122} M_{\text{Pl}}^4$ by ~ 3500 orders of magnitude.

Result 4: Multi-flavor and alternative channels do not rescue the mechanism.

- *Multi-flavor*: Inter-flavor contributions to the S/P channel vanish because $\text{tr}(\gamma^\mu) = 0$ and $\text{tr}(\gamma^\mu \gamma^5) = 0$; the Fierz traces factorize for different species [21].
- *Vector/axial condensation*: The VV and AA channels *are* attractive, but a vector condensate breaks Lorentz invariance and does not produce isotropic vacuum energy compatible with dark energy.
- *VA cross-term*: The VA Fierz coefficient vanishes identically for identical single-flavor fermions (Fierz exchange symmetry). The effective potential has $O(2)$ symmetry in (σ, π) ; Possibility B from the canonical problem statement is realized, not Possibility A [20].

4.4 Closure of Track B

Gate 1 fails on four independent lines of evidence: wrong sign, subcritical coupling, exponentially suppressed catalysis, and no multi-flavor rescue. Track B closes for the minimal single-flavor NJL pseudoscalar condensate at $\gamma = 0.274$.

Figure 1 shows G_{SP} as a function of γ , with the sign flip at $\gamma = 1$ and the LQG value $\gamma = 0.274$ marked. Figure 2 compares V_{eff} in the repulsive ($\gamma = 0.274$) and attractive ($\gamma = 2.0$) regimes.

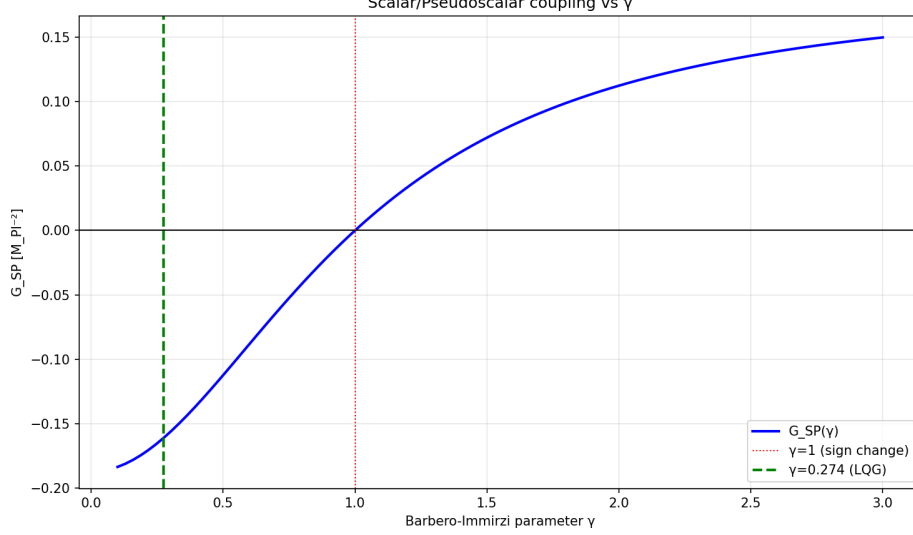


Figure 1: Effective scalar/pseudoscalar coupling G_{SP} (Eq. 6) as a function of the Barbero-Immirzi parameter γ . The coupling changes sign at $\gamma = 1$ (vertical dashed line). At the LQG value $\gamma = 0.274$ (red marker), $G_{\text{SP}} < 0$: the channel is repulsive and NJL condensation is impossible.

5 Branch G v1: Strict One-Loop Effective-Action Route

5.1 Setup

Branch G v1 asks a different question from Track B: does the one-loop gravitational effective action $\Gamma_{\text{eff}}[e]$ —obtained by integrating out all fermions on a curved background—contain a γ -dependent vacuum-like term that goes beyond standard cosmological-constant renormalization?

The target object is

$$\Gamma_{\text{eff}}[e] = S_{\text{grav}}[e, \dot{\omega}[e]] + \Gamma_{\text{ferm}}^{(1\text{-loop})}[e], \quad (9)$$

where $\Gamma_{\text{ferm}}^{(1\text{-loop})} = -\frac{1}{2} \text{Tr} \ln(-\not{D}^2 + \dots)$ is the one-loop fermion determinant.

Scope restriction. Branch G v1 is restricted to *Approach B* (strict one-loop): the four-fermion term S_{4f} is treated as a vertex in the loop expansion and does not contribute at one loop. The Hubbard-Stratonovich route (Approach A), which would reintroduce condensate-like auxiliary-field structure, is out of scope and would require a separate branch definition [17].

5.2 Gate structure

Three gates were defined and frozen before computation [17]:

Gate G1 (Nontrivial finite- γ vacuum structure)

The renormalized $\Gamma_{\text{eff}}[e]$ contains a cosmological-constant-like term whose existence depends nontrivially on finite γ and is not merely standard bare- Λ renormalization.

Gate G2 (IR physicality)

The term survives as $R \rightarrow 0$ and is scheme-robust in existence/sign.

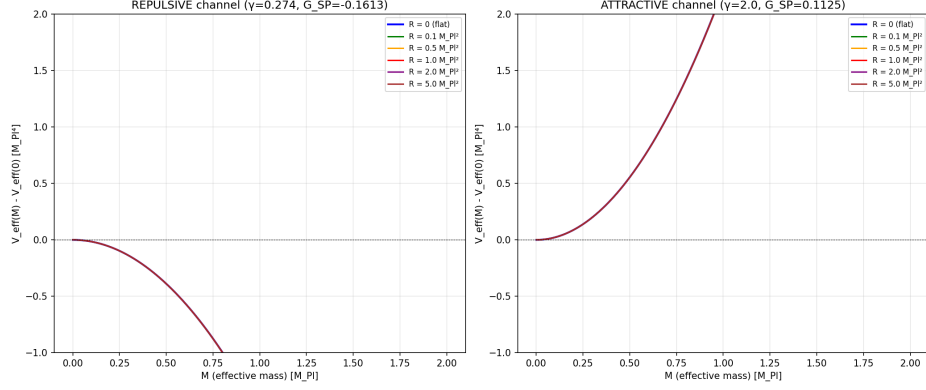


Figure 2: One-loop effective potential $V_{\text{eff}}(\pi)$ at fixed $\sigma = 0$, comparing the repulsive regime ($\gamma = 0.274$, no nontrivial minimum) with the attractive regime ($\gamma = 2.0$). Even in the attractive case, the coupling is $\sim 175\times$ below the NJL critical coupling, so no condensation occurs.

Gate G3 (Dark energy viability)

Positive vacuum energy, $|1 + w| < 10^{-2}$, mappable to the main paper's dark-energy sector.

An additional early-kill condition (FM-G6) was predeclared: if the strict one-loop object has no novel finite- γ content whatsoever, Branch G v1 closes at Phase 1 without reaching Gates G1–G3.

5.3 Structural argument

After exact torsion elimination, the reduced action is Eq. (4). The free Dirac term uses the Levi-Civita connection $\overset{\circ}{\nabla}$. All γ -dependence resides in S_{4f} , which is quartic in ψ .

The one-loop operator is γ -independent. At strict one-loop (Approach B), the Dirac operator whose determinant is computed is

$$\mathcal{D} = i\gamma^\mu \overset{\circ}{\nabla}_\mu, \quad (10)$$

the standard curved-space Dirac operator with Levi-Civita connection. This operator does not depend on γ . The four-fermion term S_{4f} is quartic in ψ and contributes only at two loops or beyond.

Therefore, $\Gamma_{\text{ferm}}^{(1\text{-loop})}[e]$ is γ -independent as a functional. Its Seeley-DeWitt coefficients $a_0, a_1, a_2, \dots$ are the standard textbook coefficients for a spin- $\frac{1}{2}$ field on a Riemannian background [6], with no γ -dependence.

Three candidate channels for γ entry. Before declaring closure, three channels through which γ might enter at one loop were checked [23]:

- (a) **Modified background equations:** The classical field equations include γ -dependent terms, so the background geometry (e.g., the bounce solution) depends on γ . However, this is γ -dependence of the *field configuration* (a point in field space), not of the effective action *as a functional*. This does not constitute novel Holst content.
- (b) **Parity-odd sector / η -invariant:** The fermion determinant phase (Atiyah-Patodi-Singer η -invariant) is computed from the spectrum of $\mathcal{D} = i\gamma^\mu \overset{\circ}{\nabla}_\mu$, which is γ -independent. Furthermore, the Pontryagin density $P_4 = 0$ on homogeneous FRW backgrounds (conformally flat),

and even on backgrounds where $P_4 \neq 0$, the contribution is curvature-local and topological, not vacuum-like.

- (c) **Functional measure corrections:** Torsion elimination is a classical algebraic substitution, not a change of fermionic integration variable. The standard treatment [2, 7] performs this without measure corrections. The path integral over ω is Gaussian in ω ; the resulting determinant factor depends on the tetrad e but not on ψ , modifying the gravitational measure rather than the fermion determinant.

Literature confirmation. Two existing one-loop computations were examined:

- Chattopadhyay (2023) [4] computes the *graviton* (not fermion) one-loop effective action in chiral EC gravity. No fermions are included; γ does not appear; the computation is for a different object.
- Shapiro & Teixeira (2014) [3] find γ -dependent one-loop divergences, but these arise from the *graviton* loop with *external* fermion currents treated as classical backgrounds. Fermions are not quantum fields in their computation. This confirms that γ enters through graviton vertices, not the fermion operator.

5.4 Closure of Branch G v1

Novel Holst content criterion (from the frozen canonical statement):

“Branch G counts as nontrivially alive only if, at the stated approximation order, the renormalized effective action $\Gamma_{\text{eff}}[e]$ contains at least one finite or regulator-robust term whose existence, sign, or classification depends nontrivially on finite γ and either disappears or changes qualitatively in the $\gamma \rightarrow \infty$ Einstein-Cartan limit.”

No such term exists at strict one-loop. FM-G6 fires. Branch G v1 closes at Phase 1 without proceeding to Gates G1–G3.

6 Route T1: Dynamical Immirzi Field

6.1 Motivation and setup

The Barbero-Immirzi parameter γ appears in the Holst action (2) as a dimensionless coupling. Promoting it to a dynamical pseudoscalar field $\gamma \rightarrow \theta(x)$ is a natural structural extension: θ then plays the role of an axion-like field whose “decay constant” is tied to the gravitational sector through the Nieh-Yan topological density [8].

The question tested in Route T1 is whether this promotion generates genuinely new gravitational content—specifically, whether the reduced action after torsion elimination contains terms that are *not* reproducible by simply adding a standard ALP to general relativity.

6.2 Literature review results

Six key references were examined [8, 9, 10, 11, 12, 13]. The assessment addressed six questions (Q1–Q6), each with predefined kill criteria.

Q1: Torsion remains algebraic. Even with $\theta(x)$ dynamical, the torsion equation of motion remains algebraic in the contorsion tensor $K^I{}_{JK}$. No derivatives of K appear. Torsion elimination proceeds exactly as in the constant- γ case [9]. This is a necessary (but not sufficient) condition for tractability.

Q2: The reduced action is a standard ALP. After torsion elimination with dynamical $\theta(x)$, the reduced action takes the form [9, 8]

$$S_{\text{reduced}}[g, \psi, \theta] = S_{\text{EH}}[g] + \int d^4x \sqrt{-g} \left[-\frac{1}{2} f(\theta) (\partial\theta)^2 + \bar{\psi} i \gamma^\mu \overset{\circ}{\nabla}_\mu \psi - m \bar{\psi} \psi + \mathcal{L}_{4f} \right], \quad (11)$$

where $f(\theta)$ is a kinetic function and $\mathcal{L}_{4f}$ is the four-fermion interaction. The critical structural result:

Every term in S_{reduced} is of a standard form. The kinetic term $f(\theta)(\partial\theta)^2$ is standard ALP kinetic energy. The Nieh-Yan coupling $\theta dT^I \wedge e_I$ reduces, after torsion elimination, to a boundary term plus standard axion-type couplings. No term arises that is not reproducible by adding a conventional pseudoscalar field to GR.

Q2 (continued): Four-fermion interaction is θ -independent. The torsion-induced four-fermion interaction after elimination is

$$\mathcal{L}_{4f} = -\frac{3}{2} \pi G (\bar{\psi} \gamma^\mu \gamma^5 \psi) (\bar{\psi} \gamma_\mu \gamma^5 \psi). \quad (12)$$

This is the standard axial-axial contact interaction and is *independent of θ* . The Fierz channel structure is unchanged from the constant- γ case analyzed in Track B. In particular, promoting $\gamma \rightarrow \theta(x)$ does not modify the scalar/pseudoscalar channel coupling or introduce new condensation channels. The Track B negative results carry over unchanged.

Q3: Literature is complete. The computation of the reduced action with dynamical Immirzi field has been performed by Calcagni & Mercuri [9] and independently by Taveras & Yunes [10]. The cosmological dynamics have been analyzed by Lattanzi & Mercuri [12]. No gap in the literature was identified.

Q4: No mechanism generates $V(\theta) \sim \Lambda$. The Mercuri [8] analogy posits that θ plays the role of the QCD axion, with the Nieh-Yan density replacing $F \wedge \tilde{F}$. However:

- In QCD, the axion potential is generated by instantons with a definite non-perturbative scale Λ_{QCD} . The resulting axion mass is $m_a \sim \Lambda_{\text{QCD}}^2 / f_a \sim 10^{-12}$ eV.
- In gravity, the analogous scale would be set by gravitational instantons, which are Planck-suppressed: $V(\theta) \sim M_{\text{Pl}}^4 e^{-S_{\text{inst}}}$ with $S_{\text{inst}} \sim M_{\text{Pl}}^2 / H^2 \gg 1$ at late times.
- The resulting vacuum energy is $\sim 10^{-36}$ orders of magnitude below $\rho_\Lambda \sim 10^{-122} M_{\text{Pl}}^4$.
- Any $V(\theta)$ yielding the observed Λ must be put in by hand. This displaces the fine-tuning problem from “why $w = -1$?” to “why $V(\theta_0) = \Lambda_{\text{obs}}$?”

Q5: Cosmological dynamics give $w = +1$, not $w = -1$. Taveras & Yunes [10] show that the dynamical Immirzi field on cosmological backgrounds behaves as stiff matter with equation of state $w = +1$. This is the *opposite* of dark energy. The field redshifts as $\rho_\theta \propto a^{-6}$, becoming negligible at late times. There is no mechanism for the field to drive late-time acceleration.

Q6: The Nieh-Yan topological nature is fatal. The Nieh-Yan 4-form $N_4 = d(e^I \wedge T_I)$ is exact (a total derivative) in four dimensions. Its topological nature ensures that any coupling of θ to N_4 reduces to boundary terms plus standard kinetic and coupling structures after torsion elimination. This is the structural reason why the gravitational origin washes out: the Nieh-Yan density carries no local dynamical information that survives torsion elimination [8, 9].

6.3 Closure of Route T1

Route T1 closes on three independent grounds:

1. **No novel content:** The reduced action after torsion elimination is identical to GR + standard ALP. Every term is reproducible without the spin-torsion origin.
2. **No vacuum mechanism:** No dynamical mechanism generates $V(\theta) \sim \Lambda$. Gravitational instantons are Planck-suppressed; any viable potential must be assumed, not derived.
3. **Wrong cosmological dynamics:** The field behaves as stiff matter ($w = +1$), the opposite of dark energy.

Route T1 was assessed through literature review only; no original symbolic computation was performed, as the existing literature fully covers the relevant calculations.

7 Route S1: Parity-Sensitive CMB Phenomenology

7.1 Motivation and setup

The ECH action contains a parity-odd operator through the Holst term $\varepsilon^{abcd} K_{ab} R_{cd}$, which sources the framework’s parity-violating vacuum structure. Cosmic birefringence—rotation of CMB polarization by an angle β —has been detected at combined significance $\beta = 0.342^\circ \pm 0.094^\circ$ (isotropic, 3.6σ) [14] with recent ACT DR6 results strengthening the evidence [15]. Route S1 asks whether the framework predicts a distinctive birefringence signal.

7.2 Assessment results

Four questions (Q1–Q4) were assessed, each with predefined kill criteria.

Q1: The minimal framework has no photon coupling. The parity-odd operator $\varepsilon^{abcd} K_{ab} R_{cd}$ is purely gravitational. It couples to the fermion spin density through torsion but has *no coupling to the photon sector* in the minimal ECH+Dirac model. The electromagnetic field strength $F_{\mu\nu}$ does not appear in the Holst term, the torsion equation of motion, or the reduced action after torsion elimination.

For the framework to produce birefringence, one must *assume* a coupling of the form $\theta F \wedge \tilde{F}$ or equivalently $g_\gamma \theta \varepsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$, with g_γ as a free parameter. This coupling is not derived from the microscopic ECH action; it must be added by hand.

The main paper itself acknowledges this gap (Sec. III.B.2 of Ref. [1]): the birefringence discussion is framed as a consistency check, not a prediction.

Q2: Parameterization is unfalsifiable. Even with an assumed photon coupling, the predicted birefringence angle is

$$\beta = f_{\text{photon}} C_0 \frac{\alpha}{M} M_{\text{Pl}}, \quad (13)$$

where f_{photon} is the assumed coupling strength (free), C_0 encodes the dilution history (free within the framework’s scaling ansatz), α/M is the framework’s dark-energy scale parameter. This formula has two free parameters (f_{photon} , C_0) for one observable (β). Any measured β can be accommodated by tuning these parameters. The mapping is unfalsifiable.

Q3: No distinguishing power from generic ALP birefringence. The predicted signal is constant- β (uniform rotation independent of multipole ℓ). This is identical to the prediction of a generic axion-like particle with constant coupling. The spin-torsion origin provides:

- no distinctive $\beta(\ell)$ shape (scale-dependent rotation),
- no cross-correlation with other observables (ΔN_{eff} , galaxy spins) that would distinguish it from a generic ALP,
- no predicted sign or magnitude constraint on β from the framework’s other fitted parameters.

All three survival criteria from the Phase 1 checklist fail.

Q4: Data exists but cannot be used distinctively. Current data (Planck NPIPE, ACT DR6) provides excellent sensitivity to isotropic birefringence [15, 14]. However, since the framework’s prediction is indistinguishable from generic constant- β ALP birefringence, fitting the data would produce a consistency check (“the framework is compatible with observed β for $f_{\text{photon}} \sim \mathcal{O}(1)$ ”), not a test of the spin-torsion mechanism.

7.3 Closure of Route S1

Route S1 closes on three independent grounds:

1. **No photon coupling:** The minimal framework has no mechanism to rotate photon polarization. Any coupling must be assumed, not derived.
2. **Unfalsifiable parameterization:** Two free parameters for one observable; any β can be accommodated.
3. **No distinguishing power:** The prediction is identical to generic ALP birefringence; no distinctive spin-torsion signature exists.

Route S1 was assessed through literature review and structural analysis; no pipeline computation was performed, as the assessment determined that no distinctive prediction exists to test.

8 Combined Interpretation

Table 1 summarizes the four tested routes and their outcomes.

The minimal ECH+Dirac model, at the tested approximation orders and extensions, does not generate the late-time vacuum term from first principles via any of the four routes tested:

- (i) the NJL condensate route (Track B),

Table 1: Summary of four tested routes to first-principles dark energy or distinctive observables in the ECH framework, and their outcomes.

Route	Target	Method	Outcome	Failure point
Track B	Condensate $\langle \bar{\psi} i \gamma^5 \psi \rangle$	Fierz + one-loop V_{eff}	Gate 1 failed	S/P channel repulsive at $\gamma = 0.274$
Branch G v1	Effective action $\Gamma_{\text{eff}}[e]$	Strict one-loop after torsion elim.	FM-G6 fired	Fermion determinant is γ -independent
Route T1	Dynamical Immirzi field $\theta(x)$	Literature review + structural analysis	All gates fail	Reduced action = standard ALP
Route S1	Parity/birefringence observables	Structural assessment	Gate S1-1 failed	No photon coupling; $\beta = \text{generic ALP}$

- (ii) the strict one-loop reduced effective-action route (Branch G v1),
- (iii) the dynamical Immirzi field extension (Route T1), or
- (iv) the parity-sensitive CMB phenomenology channel (Route S1).

Structural lesson. The four closures converge on a common structural explanation. In the minimal ECH+Dirac model, torsion is algebraic (non-propagating). Solving the torsion equation of motion exactly eliminates all Holst-sector structure, leaving only:

- the standard Einstein-Hilbert gravitational sector,
- standard minimally-coupled fermion kinetic terms, and
- a four-fermion contact interaction with constrained channel structure.

The Barbero-Immirzi parameter enters only the four-fermion coupling constants, which are subcritical for condensation (Track B) and invisible at strict one-loop (Branch G v1). Promoting $\gamma \rightarrow \theta(x)$ does not escape this structural trap: the Nieh-Yan density is topological (exact in 4D), so the dynamical field reduces to a standard ALP after torsion elimination (Route T1). And the parity-odd operator, being purely gravitational, has no photon coupling to produce distinctive birefringence (Route S1).

What this does not mean. These closures do *not* invalidate the main phenomenological framework. The main paper already treats $w = -1$ as an assumption, and these results confirm that this treatment is appropriate. The negative results establish the boundary of what the minimal model achieves at the lowest nontrivial approximation orders.

What remains open. The following routes were not tested and remain logically open, each requiring its own canonical problem statement:

- *Graviton one-loop with fermion currents* (Shapiro-Teixeira approach): γ enters through graviton vertices. This is a graviton-loop computation, structurally different from the fermion-loop computations tested here.
- *Beyond-one-loop / resummation*: The four-fermion interaction S_{4f} contributes at two loops. Hubbard-Stratonovich resummation (Approach A) may introduce γ -dependent condensate channels, but Track B showed the S/P channel fails even at this level.

- *Non-perturbative effects:* Instantons, tunneling, topological contributions.
- *Non-minimal couplings:* Additional torsion-matter interactions beyond the minimal EC+Holst+Dirac sector.
- *New symmetry structures:* Genuinely new light degrees of freedom with symmetry-protected masses, beyond the minimal model class tested here.

9 Implications and Requirements for Future Work

9.1 Status of the phenomenological framework

The phenomenological framework of Ref. [1]—the scaling ansatz $\rho_\Lambda = \Xi M_{\text{Pl}}^4$ with $w = -1$ as an assumption—is unaffected by these closures. Its cosmological consistency checks (MCMC fits to Planck/BAO/SN data), fine-tuning reduction ($10^{120} \rightarrow \sim 10^5$), and qualitative parity/birefringence consistency survive as phenomenological observations. The framework was designed to be robust under exactly this outcome.

However, the closures do establish the framework’s ceiling: it is a *motivated phenomenological model*, not a first-principles derivation of dark energy, and its minimal model class lacks both a derivation of $w = -1$ and a distinctive observational signature separating it from $\Lambda\text{CDM}+\text{ALP}$.

9.2 Requirements for next-generation models

The structural lessons from Sec. 8 translate into four concrete requirements for any candidate next-generation geometric dark-energy model:

R1: Must survive reduction without washing out.

The mechanism must not depend on degrees of freedom that are eliminated by algebraic constraint equations. If torsion is algebraic, nothing that relies on it persists in the reduced action.

R2: Must address tiny-scale naturalness.

The model must explain why the relevant scale is $\sim 10^{-122} M_{\text{Pl}}^4$ through a structural mechanism (symmetry, sequestering, relaxation), not merely by setting a free parameter to the observed value.

R3: Must offer a distinctive prediction.

The model must produce at least one observable or mechanism that is not reproducible by ΛCDM plus a generic axion-like particle.

R4: Must fail cleanly if wrong.

The model must be testable with predefined gates and kill criteria, producing documentable negative results if it fails.

9.3 Candidate directions beyond the minimal model

Three directions satisfy at least one of these requirements while addressing at least one structural lesson:

- *Propagating torsion* (Poincaré gauge theory): torsion becomes dynamical, avoiding the algebraic wash-out (R1). Risk: ghost instabilities in the massive torsion sector.

- *Symmetry-protected geometric pseudoscalars*: a pseudoscalar with a protected mass and non-topological geometric coupling, avoiding ALP collapse (R3) and addressing naturalness (R2). Risk: may reduce to a standard ALP in disguise.
- *Vacuum sequestering*: global constraints enforce radiative stability of Λ , addressing naturalness directly (R2). Risk: Weinberg no-go theorem; potentially no distinctive observable (R3).

Each requires its own canonical problem statement and gate structure before computation.

10 Reproducibility and Repository Layout

Track B and Branch G v1 computations are symbolic or semi-analytic, implemented in Python 3.14 with SymPy 1.14, NumPy, SciPy, and Matplotlib. No GPU or specialized hardware is required. Routes T1 and S1 are literature-review and structural-assessment closures with no original computation; their documentation is archived in the repository. The full repository is organized as follows.

10.1 Frozen documents (not to be edited)

File	Content
06_canonical_problem_statement.md	Track B canonical problem statement (v3, frozen). Defines 7 questions, 3 gates, order parameter, symmetry logic, regulator strategy.
06a_frozen_assumptions_and_change_log.md	Track B freeze log with reopening criteria and v1→v2→v3 change history.
07_track_b_closure.md	Track B closure document. Four load-bearing negative results, artifact inventory, impact assessment.
08_branchG_canonical_problem_statement.md	Branch G v1 canonical problem statement (v2, frozen). 6 questions, 3 gates, FM-G6 early kill, scope control.
08a_branchG_freeze_log.md	Branch G v1 freeze log with Approach A/B label clarification and Track B comparison table.

10.2 Computation scripts

Script	Type	Content
derivations/comp1_torsion_elimination.py	Symbolic	Torsion elimination in ECH gravity. Derives perfect-square structure (5), coupling constants, $\gamma \rightarrow \infty$ limit check.
derivations/comp2_fierz_rearrangement.py	Symbolic	Full 16×16 Fierz rearrangement using explicit gamma matrices in the Dirac representation. Channel decomposition.

<code>derivations/comp2b_fierz_verification.py</code>	Symbolic	Independent verification of Fierz results. VA vanishing proof, sign analysis, Fierz completeness check.
<code>derivations/comp3_one_loop_veff.py</code>	Semi-analytic	One-loop V_{eff} with curvature corrections. γ -scan for nontrivial minima (none found). Multi-flavor analysis. Generates Figs. 1–2.
<code>derivations/comp3b_gap_equation_checker.py</code>	Numerical	NJL gap equation in proper-time regularization. Critical coupling comparison ($G_{\text{SP}}/G_{\text{crit}} \approx 0.006$). Curvature catalysis scan.
<code>derivations/branchG_phase1_operator_analysis.py</code>	Analysis	Branch G Phase 1: three-channel analysis of whether γ enters the strict one-loop Dirac operator.

10.3 Analysis notes and status reports

File	Content
<code>notes/gate1_status_20260313.md</code>	Gate 1 failure report: three results that kill the S/P condensate route. Remaining active branches.
<code>notes/comp3_results_analysis.md</code>	Fierz result analysis: VA cross-term structure, multi-flavor implications, Possibility A vs B resolution.
<code>notes/branchG_phase1_closure.md</code>	Branch G Phase 1 closure: FM-G6 declaration, three-channel verdict, literature comparison.

10.4 Route T1 and S1 documentation

Routes T1 and S1 were closed through literature review and structural assessment. No original symbolic computation was required. The following documents record the assessments:

File	Content
<code>next_branches/T1_dynamical_immirzi/01_canonical_problem_statement.md</code>	T1 canonical problem statement. 6 questions, 3 gates, 6 failure modes, 4-phase computation sequence.
<code>next_branches/T1_dynamical_immirzi/06_phase1_literature_review_questions.md</code>	T1 Phase 1 checklist: 6 sharp questions with kill criteria, 7-day reading plan, assessment memo template.
<code>next_branches/S1_parity_cmb/01_canonical_problem_statement.md</code>	S1 canonical problem statement. 4 target observables, 3 gates, 5 failure modes.
<code>next_branches/S1_parity_cmb/05_phase1_assessment_checklist.md</code>	S1 Phase 1 checklist: 4 questions with kill criteria, 5-day timeline, mapping memo template.
<code>next_branches/01_sequencing_and_decision_gates.md</code>	Master sequencing table, gate dependencies, critical path analysis for all four branches (T1, T2, S1, S2).

10.5 Figures

File	Content	
figures/G_SP_vs_gamma.png	G_{SP} vs. γ plot (Fig. 1).	Generated by <code>comp3_one_loop_veff.py</code> .
figures/veff_repulsive_vs_attractive.png	Long comparison plot (Fig. 2).	Generated by <code>comp3_one_loop_veff.py</code> .

10.6 Reproduction instructions

To reproduce all results:

1. Install Python 3.14 with `pip install sympy numpy scipy matplotlib`.
2. Navigate to `ir_vacuum_program/derivations/`.
3. Run scripts in order: `comp1_torsion_elimination.py` $\rightarrow$ `comp2_fierz_rearrangement.py` $\rightarrow$ `comp2b_fierz_verification.py` $\rightarrow$ `comp3_one_loop_veff.py` $\rightarrow$ `comp3b_gap_equation_check.py` $\rightarrow$ `branchG_phase1_operator_analysis.py`.
4. All output is printed to stdout. Figures are written to `../figures/`.
5. No script depends on external data, GPU, or network access.

10.7 What is symbolic vs. numerical

Category	Scripts
Purely symbolic (SymPy)	<code>comp1</code> , <code>comp2</code> , <code>comp2b</code> : gamma matrix algebra, Fierz traces, coupling constants. Exact algebraic results.
Semi-analytic	<code>comp3</code> : symbolic setup + numerical γ -scan of V_{eff} . <code>comp3b</code> : analytic gap equation + numerical root-finding and catalysis estimates.
Structural/analytic	<code>branchG_phase1</code> : logical argument verified against literature; no numerics beyond illustrative coefficient values.

11 Conclusion

Four routes to deriving the late-time vacuum term or identifying distinctive observables in the spin-torsion dark-energy framework were tested and found to fail:

1. The NJL condensate route (Track B) fails because the scalar/pseudoscalar channel is repulsive at the LQG Barbero-Immirzi value $\gamma = 0.274$, the gravitational four-fermion coupling is $\sim 175\times$ subcritical, and gravitational catalysis is exponentially suppressed.
2. The strict one-loop effective-action route (Branch G v1) fails because the fermion determinant after torsion elimination is the standard γ -independent curved-space Dirac determinant, with no novel Holst content at this order.

3. The dynamical Immirzi field extension (Route T1) fails because torsion elimination reduces the dynamical $\theta(x)$ field to a standard axion-like particle with no novel gravitational content, no mechanism for generating $V(\theta) \sim \Lambda$, and cosmological dynamics yielding $w = +1$ (stiff matter).
4. The parity-sensitive CMB phenomenology assessment (Route S1) fails because the minimal framework has no photon coupling, the assumed coupling produces constant- β birefringence identical to generic ALP phenomenology, and the parameterization is unfalsifiable.

These closures share a common structural origin: in the minimal ECH+Dirac model, torsion elimination removes all Holst-sector structure, and the Nieh-Yan topological density carries no local dynamical content that survives this elimination. The resulting physics is standard GR + fermions + a constrained four-fermion interaction, with no mechanism for late-time vacuum energy and no distinctive observational signature.

These are clean negative results with a unifying structural explanation: algebraic torsion in the minimal ECH model washes out completely after elimination, and the Nieh-Yan density’s topological nature in four dimensions ensures that even the simplest structural extension (dynamical Immirzi field) collapses to standard physics.

The value of these results is threefold. First, they map the landscape of what does not work within the most natural minimal model, preventing duplication of effort. Second, they identify the precise structural bottleneck—algebraic torsion and topological Nieh-Yan—that any successor model must overcome. Third, they establish a reproducible methodology (canonical problem statements, predeclared gates, frozen scope) for disciplined model testing that can be applied to future candidates.

The phenomenological framework of Ref. [1] survives as a motivated ansatz; the first-principles derivation of $w = -1$ and the identification of a distinctive observable remain open problems requiring physics beyond the minimal model class.

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